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(57) Abstract:

Efficient water management and sustainable agricultural practices remain critical challenges in modem farming, particularly due to unpredictable climatic conditions, inefficient irrigation methods, and lack of real-time monitoring systems. Traditional irrigation techniques are often manual, resource-intensive, and unable to adapt dynamically to changing environmental conditions, resulting in water wastage, reduced crop productivity, and soil degradation. The proposed system, An loT-Enabled Intelligent System for Adaptive Irrigation and Rainwater Harvesting Using Machine Learning, introduces an advanced smart agriculture solution that integrates Internet of Things (IoT) sensors, machine learning using Gradient Boosting algorithms, and real-time data analytics to optimize irrigation and improve water resource utilization. The system incorporates environmental sensors such as soil moisture, temperature, humidity, and rainfall detection, along with ESP Cam-based visual monitoring and communication modules for continuous field monitoring and remote access. By analysing multi-parameter data, the machine learning model predicts optimal irrigation schedules and water requirements, enabling automated drip irrigation and intelligent rainwater harvesting. The system dynamically regulates water distribution, minimizes wastage, and enhances crop health and yield. Additionally, real-time alerts and monitoring dashboards support informed decision-making while reducing manual intervention. Overall, the proposed solution provides a scalable, cost-effective, and data- driven framework for precision agriculture, promoting sustainable farming practices and efficient water management across diverse agricultural environments.

FORM 2

THE PATENT ACT 1970
(39 of 1970)
&
The Patent Rules, 2003
PROVISIONAL/COMPLETE SPECIFICATION
(See section 10 and rule 13)

710170447

1. **TITLE OF THE INVENTION**: An IoT-Enabled Intelligent System for Adaptive Irrigation and Rainwater Harvesting Using Machine Learning

2. **APPLICANT(S):**

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3. **PREAMBLE TO THE DESCRIPTION**

COMPLETE

The following specification describes the invention and the manner in which it is to be performed:

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ABSTRACT OF THE INVENTION

Efficient water management and sustainable agricultural practices remain critical challenges in modern farming, particularly due to unpredictable climatic conditions, inefficient irrigation methods, and lack of real-time monitoring systems. Traditional irrigation techniques are often manual, resource-intensive, and unable to adapt dynamically to changing environmental conditions, resulting in water wastage, reduced crop productivity, and soil degradation. The proposed system, An IoT-Enabled Intelligent System for Adaptive Irrigation and Rainwater Harvesting Using Machine Learning, introduces an advanced smart agriculture solution that integrates Internet of Things (IoT) sensors, machine learning using Gradient Boosting algorithms, and real-time data analytics to optimize irrigation and improve water resource utilization. The system incorporates environmental sensors such as soil moisture, temperature, humidity, and rainfall detection, along with ESP Cam-based visual monitoring and communication modules for continuous field monitoring and remote access. By analysing multi-parameter data, the machine learning model predicts optimal irrigation schedules and water requirements, enabling automated drip irrigation and intelligent rainwater harvesting. The system dynamically regulates water distribution, minimizes wastage, and enhances crop health and yield. Additionally, real-time alerts and monitoring dashboards support informed decision-making while reducing manual intervention. Overall, the proposed solution provides a scalable, cost-effective, and data-driven framework for precision agriculture, promoting sustainable farming practices and efficient water management across diverse agricultural environments.

DATE AND SIGNATURE

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CLAIMS

We Claim:

1. An IoT-enabled intelligent system for adaptive irrigation and rainwater harvesting using machine learning, comprising a network of environmental sensors, a processing unit, and an automated irrigation mechanism, wherein the system continuously monitors real-time agricultural parameters to optimize water usage and improve crop productivity.
2. The system as claimed in Claim 1, wherein the environmental sensors include soil moisture, temperature, humidity, rainfall, and water level sensors configured to collect multi-parameter field data for analysis.
3. The system as claimed in Claim 1, wherein the machine learning model utilizes a Gradient Boosting algorithm to analyse sensor data and predict optimal irrigation schedules and water requirements based on environmental and crop conditions.
4. The system as claimed in Claim 1, wherein the irrigation mechanism comprises an automated drip irrigation system controlled through actuators and solenoid valves for precise water distribution.
5. The system as claimed in Claim 1, further comprising an integrated rainwater harvesting module configured to collect, store, monitor, and distribute harvested rainwater intelligently.
6. The system as claimed in Claim 1, further comprising an ESP Cam module for real-time visual monitoring of crop conditions, enabling remote surveillance and decision support.
7. The system as claimed in Claim 1, wherein communication modules including GSM and/or wireless protocols are used for transmitting sensor data and system alerts to a remote interface.
8. The system as claimed in Claim 1, further comprising a cloud-based or server-based backend with database storage for maintaining historical data, model outputs, and system logs, and supporting predictive analytics along with a user interface dashboard for real-time monitoring and control.
9. The system as claimed in Claims 1 through 8, wherein the integration of IoT, machine learning, automated irrigation, and rainwater harvesting enables efficient water resource utilization, reduced manual intervention, and improved crop yield and sustainability.

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4. DESCRIPTION

FIELD OF THE INVENTION

The present disclosure relates to the field of smart agriculture and precision farming, and more particularly to an IoT-enabled intelligent system integrated with machine learning for adaptive irrigation control, rainwater harvesting, and real-time environmental monitoring to optimize water usage and enhance crop productivity.

INTRODUCTION

Agriculture remains highly dependent on water availability and environmental conditions, making efficient water management a critical factor in ensuring sustainable crop production. Conventional irrigation practices are often manual, inefficient, and lack adaptability to real-time field conditions, leading to excessive water consumption, uneven distribution, and reduced crop yield. Additionally, the absence of continuous monitoring and predictive decision-making mechanisms limits the ability of farmers to respond effectively to changing climatic conditions.

With advancements in Internet of Things (IoT) and Artificial Intelligence (AI), smart agriculture systems have emerged as a promising solution to address these challenges. The proposed system integrates multiple environmental sensors, machine learning algorithms, and automated control mechanisms to enable real-time monitoring and intelligent decision-making. By leveraging a Gradient Boosting-based predictive model, the system analyses environmental and soil parameters to determine optimal irrigation schedules. Furthermore, the inclusion of ESP Cam-based visual monitoring enhances field surveillance, while rainwater harvesting mechanisms ensure efficient utilization of natural resources. The system also supports remote monitoring through communication modules and user dashboards, thereby reducing manual intervention and improving operational efficiency. Overall, the integration of IoT and AI transforms traditional farming into a data-driven, sustainable, and efficient agricultural practice.

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OBJECTIVES OF THE INVENTION

The objectives of the invention are:

- To develop an IoT-based intelligent system for real-time monitoring of environmental and soil conditions.
- To implement a machine learning model using Gradient Boosting for accurate prediction of irrigation requirements.
- To design an automated irrigation system that optimizes water distribution based on real-time and predictive data.
- To integrate a rainwater harvesting mechanism for efficient collection, storage, and utilization of natural water resources.
- To incorporate ESP Cam-based visual monitoring for enhanced crop surveillance and field analysis.
- To enable remote monitoring and control through communication modules and user-friendly dashboards.
- To minimize water wastage and improve water efficiency in agricultural practices.
- To enhance crop health, yield, and overall farm productivity through data-driven decision-making.
- To provide a scalable and cost-effective solution adaptable to diverse agricultural environments.

SUMMARY OF THE INVENTION

- The invention presents an IoT-enabled smart agriculture system that integrates real-time sensing, machine learning, and automated control for efficient irrigation and water management.
- The system utilizes multiple environmental sensors and a Gradient Boosting-based predictive model to determine optimal irrigation schedules and water requirements.
- It incorporates automated drip irrigation and rainwater harvesting mechanisms to ensure efficient water utilization and sustainability.
- The integration of ESP Cam-based monitoring, remote dashboards, and alert systems enhances decision-making and reduces manual effort.

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- Overall, the system provides a scalable, cost-effective, and intelligent framework for precision agriculture, improving crop productivity and resource efficiency.

WORKING MODEL

The proposed invention operates through an integrated workflow combining IoT sensing, data processing, machine learning prediction, and automated actuation. Initially, environmental and soil parameters such as moisture, temperature, humidity, and rainfall are continuously collected using distributed sensors deployed across the agricultural field. The collected data is transmitted to a central processing unit or cloud-based server through communication modules.

The machine learning model processes the incoming data to predict optimal irrigation timing and water quantity required for specific crop conditions. Based on these predictions, control signals are sent to the irrigation system, which activates drip irrigation through valves and pumps. Simultaneously, the rainwater harvesting module collects and stores rainwater, which is later utilized as an alternative water source.

The ESP Cam module provides real-time visual monitoring of crop conditions, enabling users to remotely observe field status. All system data, predictions, and alerts are displayed on a user interface dashboard, allowing farmers to monitor, control, and make informed decisions efficiently. This closed-loop system ensures adaptive irrigation, efficient water utilization, and improved crop outcomes.

MODULE DESCRIPTION

Module 1 (Soil and Environmental Monitoring)

This module is responsible for collecting real-time data from various sensors, including soil moisture, temperature, humidity, and rainfall sensors. The data serves as the primary input for analysis and decision-making, enabling accurate assessment of field conditions.

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Module 2 (Machine Learning-Based Prediction)

This module utilizes a Gradient Boosting algorithm to analyse sensor data and predict optimal irrigation schedules and water requirements. The model is trained on multiple environmental parameters to ensure high accuracy and adaptability to varying conditions.

Module 3 (Smart Irrigation Control)

The smart irrigation module automates water distribution using drip irrigation systems controlled by actuators and solenoid valves. Based on model predictions, it ensures precise and efficient water delivery, minimizing wastage and maintaining optimal soil moisture levels.

Module 4 (Rainwater Harvesting System)

This module collects and stores rainwater using tanks and monitoring sensors. It includes water level detection and controlled distribution mechanisms to efficiently utilize harvested rainwater for irrigation purposes.

Module 5 (Visual Monitoring using ESP Cam)

The ESP Cam module enables real-time visual surveillance of the agricultural field. It assists in monitoring crop conditions, detecting anomalies, and providing additional insights beyond sensor data.

Module 6 (Communication and Remote Access)

This module includes GSM and/or wireless communication technologies to transmit data between the field system and the user interface. It enables remote monitoring, control, and alert notifications for efficient farm management.

Module 7 (User Dashboard and Data Management)

This module provides a user-friendly interface for visualizing real-time data, system status, alerts, and predictions. It also stores historical data in a database, supporting analytics, reporting, and continuous system improvement.

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DRAWINGS

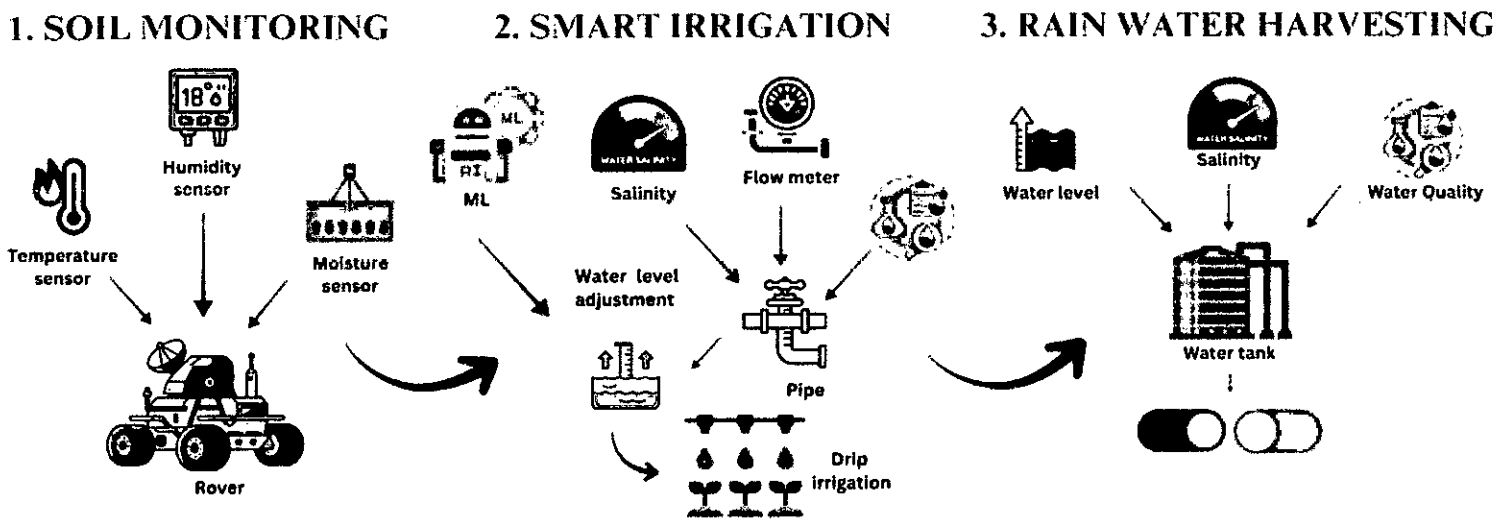


Figure 1: Architecture diagram of An IoT-Enabled Intelligent System for Adaptive Irrigation and Rainwater Harvesting Using Machine Learning

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Total No. of Sheets: 5
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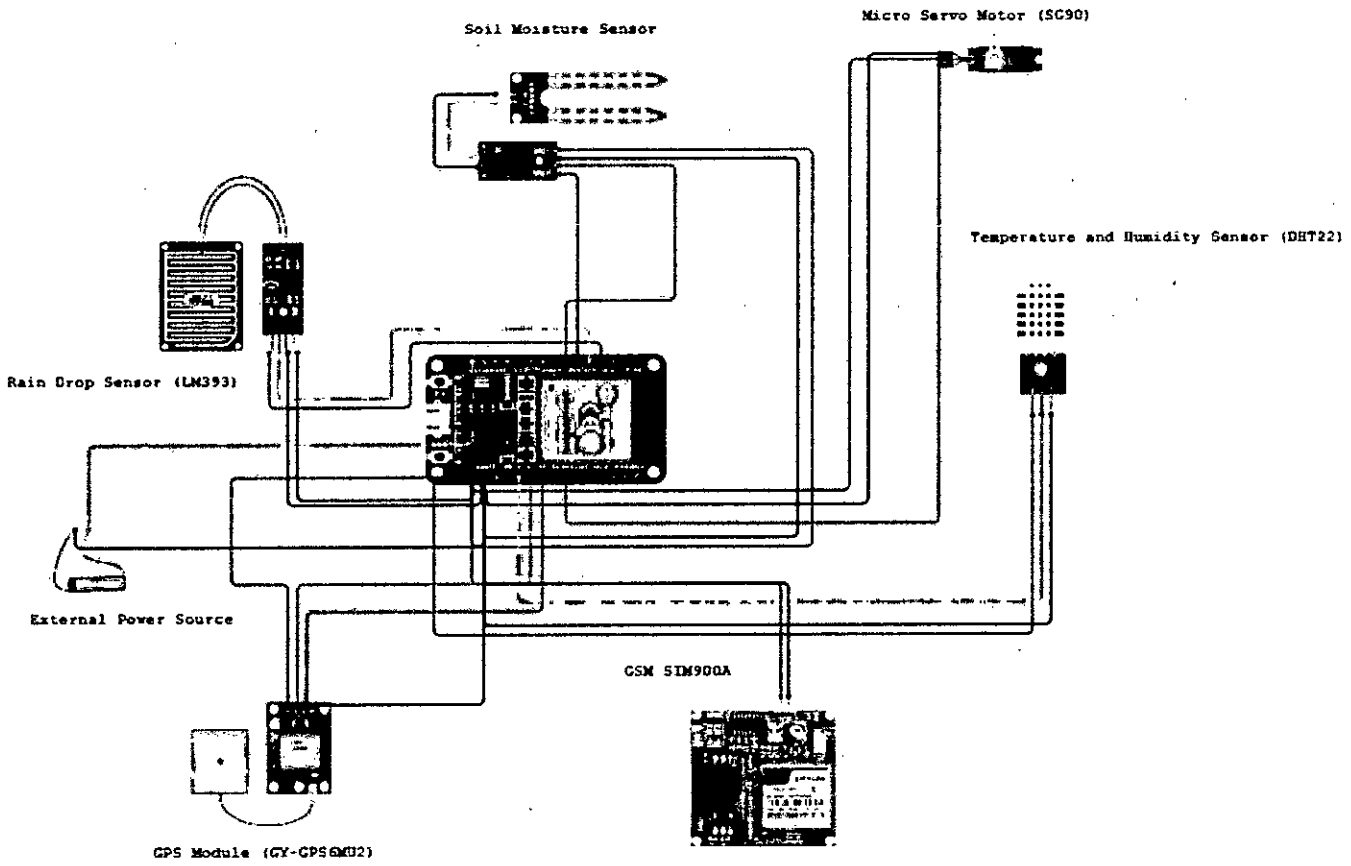


Figure 2: Circuit diagram of An IoT-Enabled Intelligent System for Adaptive Irrigation and Rainwater Harvesting Using Machine Learning

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Total No. of Sheets: 5
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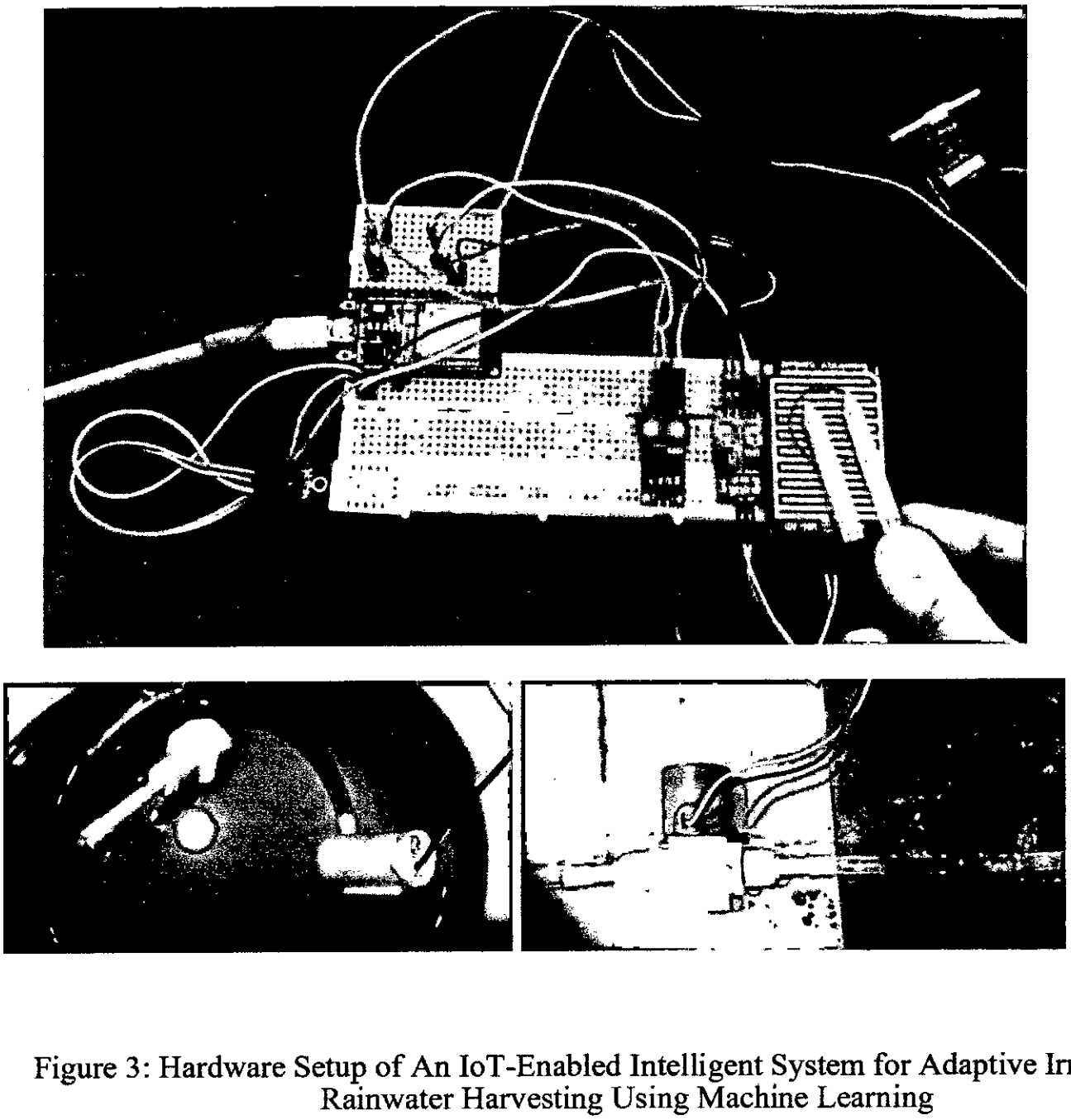


Figure 3: Hardware Setup of An IoT-Enabled Intelligent System for Adaptive Irrigation and Rainwater Harvesting Using Machine Learning

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Total No. of Sheets: 5
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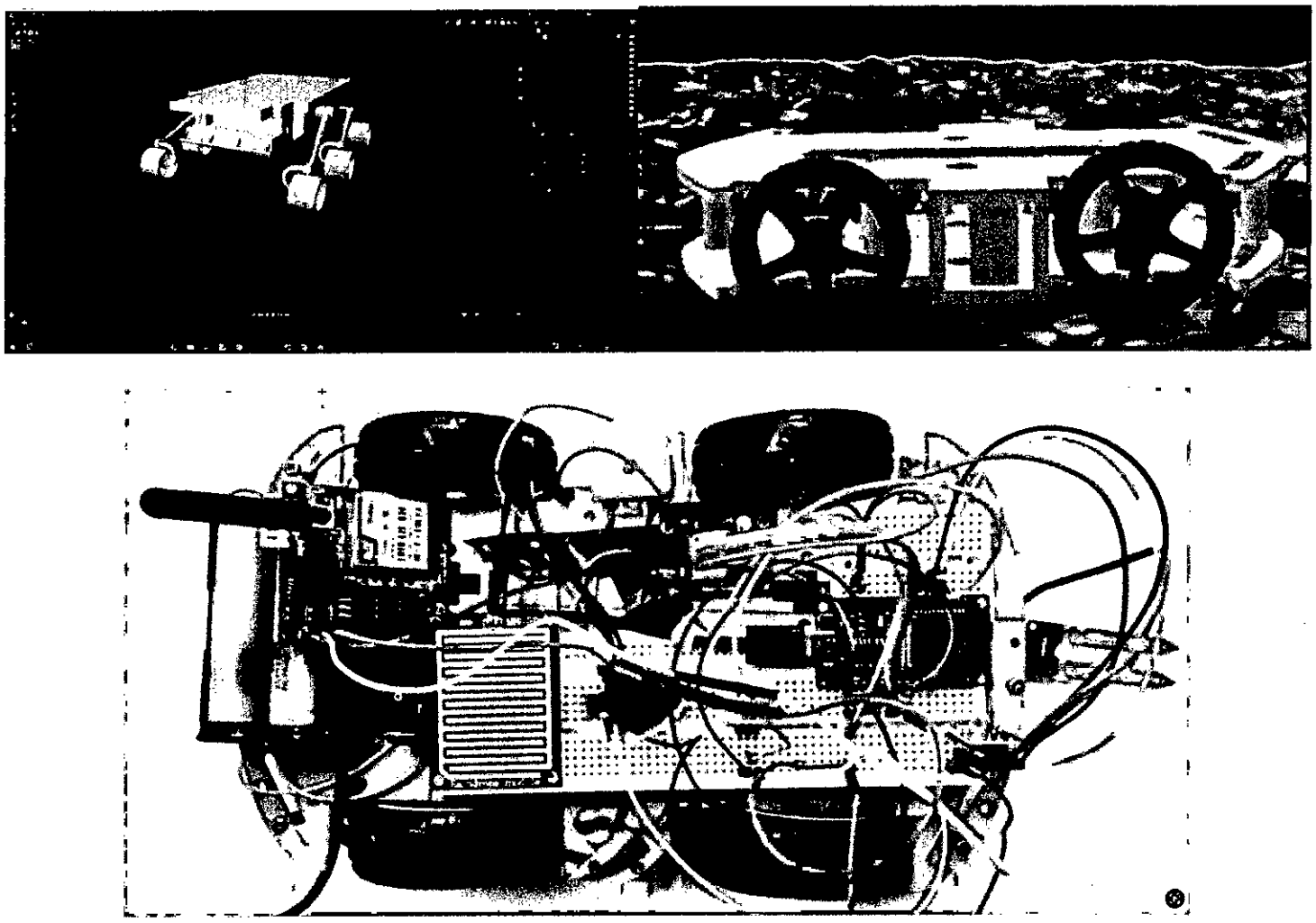


Figure 4: 3D Modelling, Simulation and Rover Built for An IoT-Enabled Intelligent System for Adaptive Irrigation and Rainwater Harvesting Using Machine Learning

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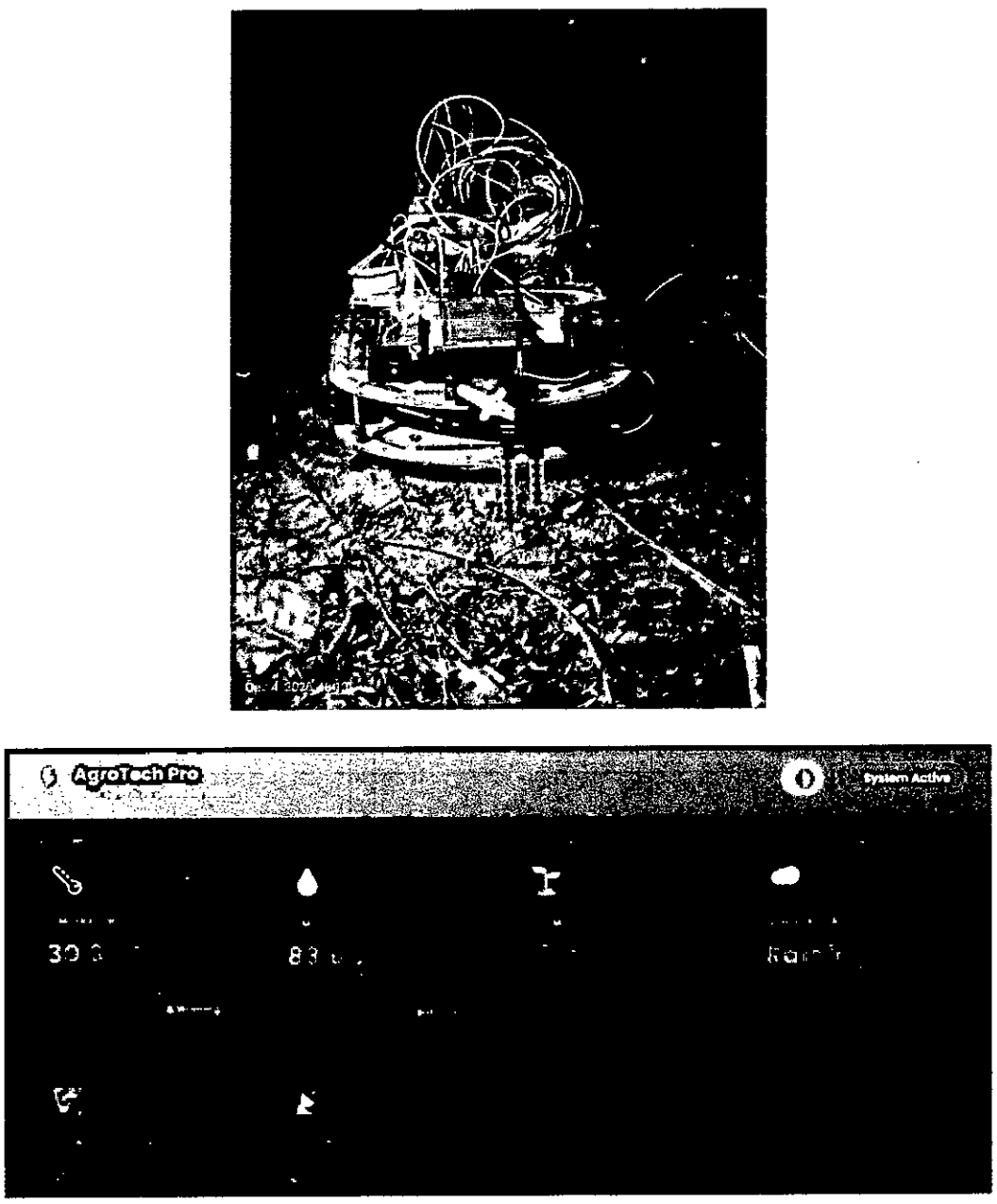


Figure 5: Real-time Testing and Output of An IoT-Enabled Intelligent System for Adaptive Irrigation and Rainwater Harvesting Using Machine Learning

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